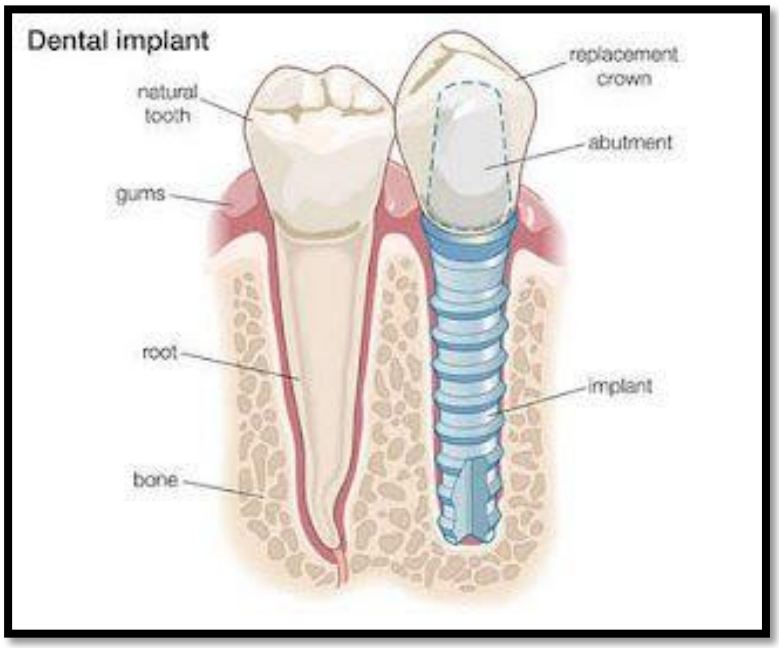
Materials for construction of implant

Dental implants are fixtures that serve as replacements for the root of a missing natural tooth. Implants may be placed in the mandible or maxilla. When properly designed and placed, dental implants bond with bone over time and serve as an anchor for dental prostheses. Dental implants are used to replace a single missing tooth or many teeth, or to support a complete removable denture



Osseointegration was initially defined on the light microscopic level as "a direct structural and functional connection between ordered, living bone and the surface of a load-carrying implant

CLASSIFICATION

Dental implants fall into 1 of the following 3 primary groups: *(1)* metals, *(2)* ceramics, and *(3)* polymers. In addition, biomaterials can be classified based on the

type of biologic (tissue) response when implanted and the interaction that develops with the host tissue (bone) . Three major types of dental implants : *(1)* biotolerant, *(2)* bioinert, and *(3)* bioactive

Bioinert Biomaterials

- The term bioinert refers to any material that once placed in the human body has minimal interaction with its surrounding tissue, examples of these are stainless steel, titanium, alumina, zirconia, and ultra high molecular weight polyethylene.
- Bioinert materials allow close apposition of bone on their surface, leading to contact osteogenesis.

Bioactive Biomaterials

- Bioactive refers to a material, which upon being placed within the human body interacts with the surrounding bone that is chemically equivalent to the mineral phase in bone.
- examples of these materials are synthetic hydroxyapatite [$Ca_{10}(PO_4)_6(OH)_2$], glass ceramic and bioglass

- Bioactive materials also allow the formation of new bone, active carbonate apatite (CHAp) layer on the implant by ion exchange with host tissue leads to the formation of a chemical bond along the interface (bonding osteogenesis)

Biotolerant

- Gold ,Co-Cr alloys , Polyethylene ,Polyamide Polymethylmethacrylate
- Biotolerant materials are those that are not necessarily rejected when implanted into living tissue, but are surrounded by a fibrous layer in the form of a capsule

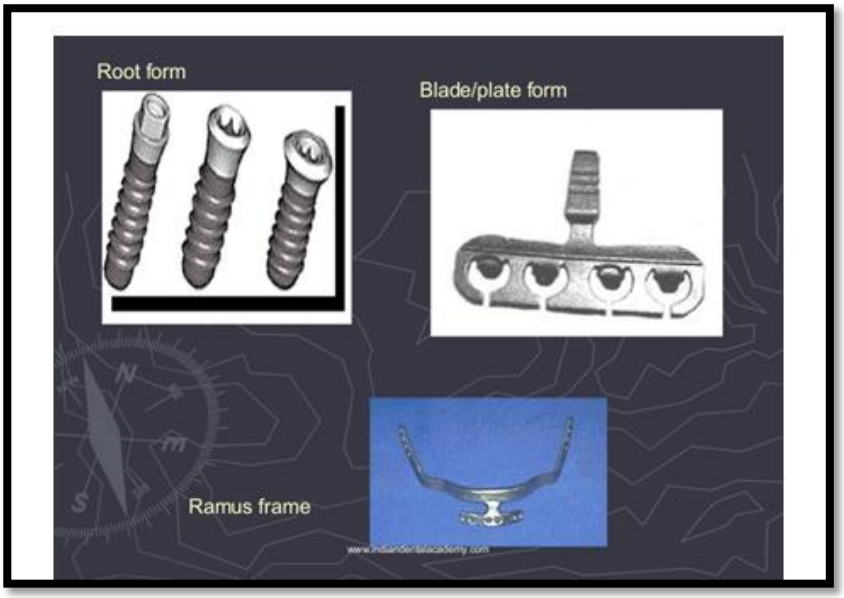
Historically, dental implants have been classified according to their design and placement within the tissues. The three types of implants commonly used are :

The subperiosteal implant, the transosteal implant, and the endosseous implant .

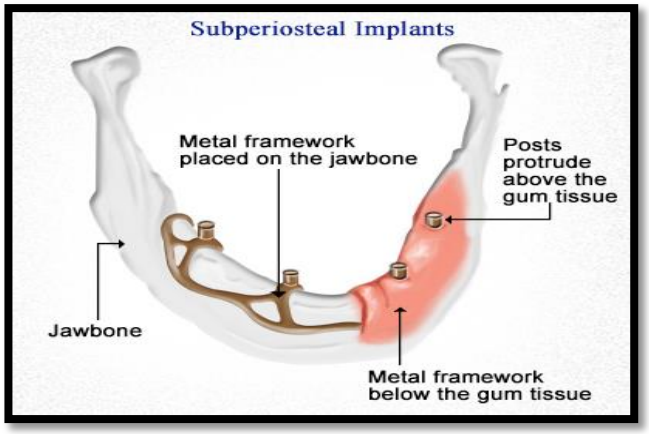
Endosseous Implant

Endosseous implants are the most common type of implant placed today. Implants are placed directly into the mandible or maxilla

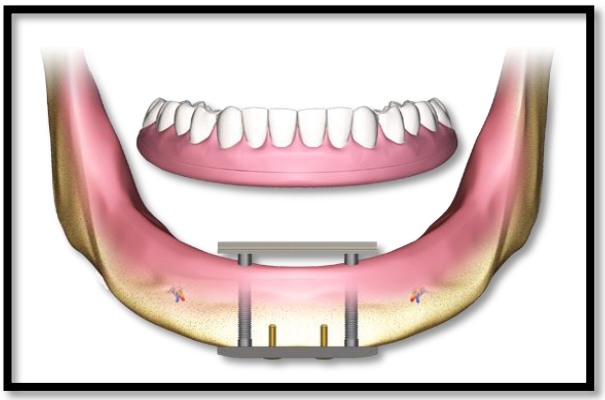
Endosseous implant



Subperiosteal implant: is a type of implant where the artificial tooth is placed beneath the periosteum that overlies the cortex.



Transosteal implant: This type of implant can, also, be referred to as staple bone implant, mandibular staple implant or transmandibular implant. This type of implant is a combination of both the endosteal components and those of subperiosteal. The implant penetrates the two cortical plates.



Properties of an implant biomaterial

Modulus of elasticity: Implant material with modulus of elasticity comparable to bone (18 GPa) must be selected to ensure more uniform distribution of stress at implant and to minimize the relative movement at implant bone interface.

Metals possess high strength and ductility, whereas the ceramics and carbons are brittle materials

Tensile, compressive and shear strength: An implant material should have high tensile and compressive strength to prevent fractures and improve functional stability. Improved stress transfer from the implant to bone is reported interfacial shear strength is increased, and lower stresses in the implant.

Yield strength, fatigue strength: An implant material should have high yield strength and fatigue strength to prevent brittle fracture under cyclic loading.

Ductility: According to ADA a minimum ductility of 8% is required for dental implant. Ductility in implant is necessary for contouring and shaping of an implant.

Hardness and Toughness: Increase in hardness decreases the incidence of wear of implant material and increase in toughness prevents fracture of the implants.

Metallic implants

Titanium

• Bioinert

•Lightweight

•biocompatible

•corrosion resistant (dynamic inert oxide layer , TiO_2)

•It is 6 times stronger than compact bone

•Its modulus of elasticity is 5 times greater than that of compact bone (102-110 GPs) (thus equal mechanical stress transfer)

•Most commonly used –Commercially pure (CP) titanium

–Titanium-aluminum-vanadium alloy (Ti-6Al-4V) -

stronger & used with smaller diameter implants

Cobalt-chromium-molybdenum alloys generally consist of 63% cobalt, 30% chromium, and 5% molybdenum with small amounts of carbon, manganese, and nickel. Molybdenum is a stabilizer and provides strength; chromium provides the passivating effect to ensure corrosion resistance through the oxide surface; and carbon serves as a hardener.

Co-Cr-Mo alloys have a high elastic modulus and resistance to corrosion.

Co-Cr-Mo and stainless steel alloy continue to be used for some implants, such as subperiosteal and transosteal implants and ramus frames, because of their castability, mechanical properties, and lower cost.

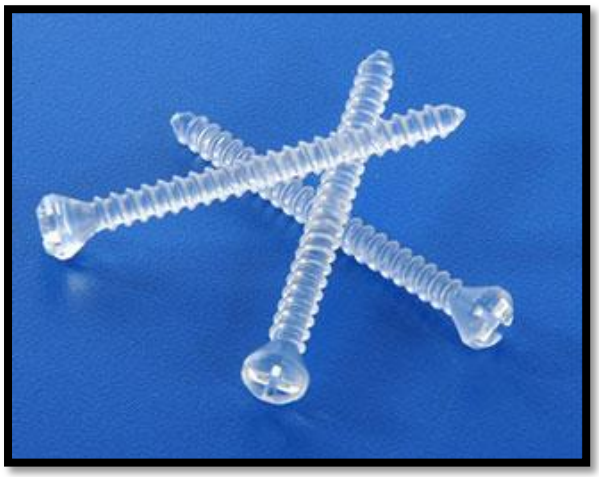
Ceramic implant

Ceramics are inorganic, nonmetallic materials . Ceramic implants can withstand only relatively low tensile or shear stresses induced by occlusal loads, but they can tolerate quite high levels of compressive stress. Aluminum oxide (Al_2O_3) is used as a standard biomaterial for ceramic implants because of its inertness (biostability).

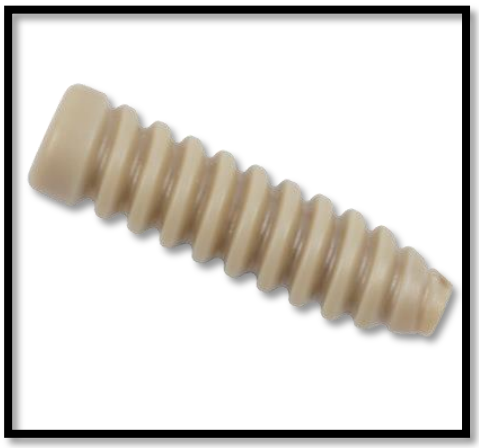
Zirconia (ZrO_2) has also demonstrated a high degree of inertness.
These types of ceramic implants are not bioactive in that they do not promote the formation of bone. They have high strength, stiffness, and hardness and function very well for some designs of dental implants
Hydroxyapatite (HA), or $Ca_{10}(PO_4)_6(OH)_2$, and tricalcium phosphate (TCP), $Ca_3(PO_4)_2$ promote and achieve a direct bond of the implant to hard tissues, they are classified as bioactive.

POLYMERS

The early work with the methyl methacrylate resin implants met mostly with failures .
There are some disadvantages: (1) inferior mechanical properties; (2) lack of adhesion to living tissues; and (3) adverse immunologic reactions.



Peek (Polyetheretherketone) as dental implants

The major beneficial property for peek is its low elastic modulus (3-4 GPs) being close to human bone , good resistance to degradation, lack of toxicity, good chemical and sterilization resistance, lighter in weight,

PEEK combines high strength with a relatively low Young's modulus which is closer to that of human bone than titanium. This property may minimize the stress by distributing it in more physiological manner thus supporting bone formation around the implant and reducing osteolysis.